Advanced Fluid Dynamics

DAY 7

Convection problem in OpenFOAM

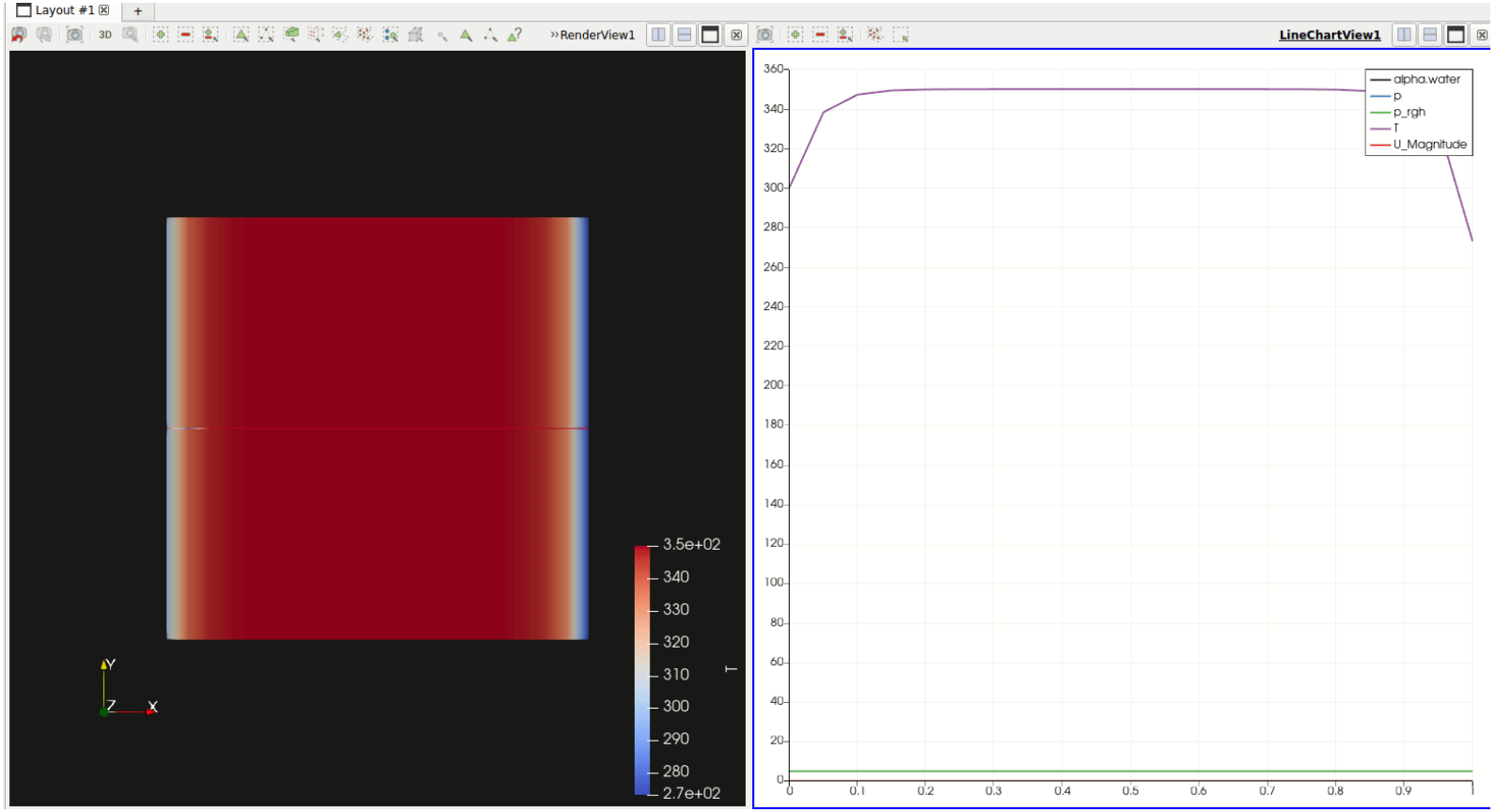
Kumaresh

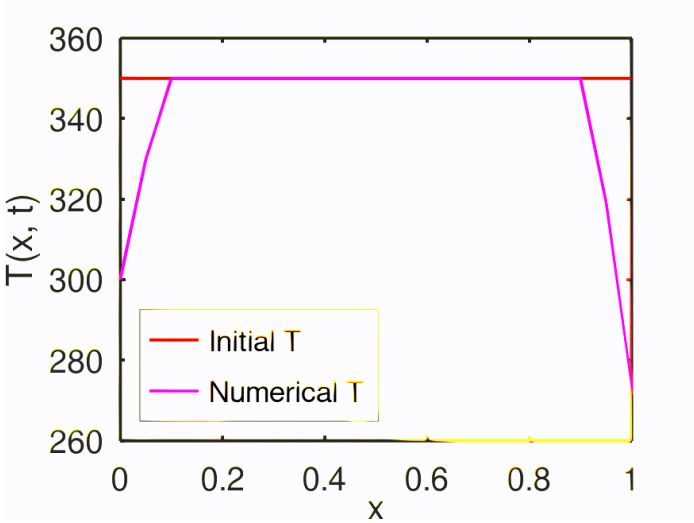
Contents

- Explanation about Convection equation
- Project – 1 (Last week)
- Project – 2

OpenFOAM: Numerical Solution to Diffusion Equation

Paraview – Post Processing tool in OpenFOAM





$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Extract files from GitHub

AdvancedFluidDynamics / DAY6-OF_DiffusionEqu /

Kumaresh1176 Add files via upload

Name	Last commit message
..	
DAY6-OpenFOAM_diffusion_equation.rar	Add files via upload
DAY6_AdvancedFluidDynamics.pdf	Add files via upload
Readme.md	Create Readme.md

Readme.md

Project 1: #7

solver file ➔ diffusionFoam/createFields.H

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Code | Blame | 49 lines (43 loc) · 846 Bytes | Code 55% faster with GitHub Copilot

```
Info<< "Reading field T\n" << endl;

volScalarField T
(
    IOobject
    (
        "T",
        runTime.timeName(),
        mesh,
        IOobject::MUST_READ,
        IOobject::AUTO_WRITE
    ),
    mesh
);


Info<< "Reading diffusivity DT\n" << endl;

volScalarField DT
(
    IOobject
    (
        "DT",
        runTime.timeName(),
        mesh,
        IOobject::READ_IF_PRESENT,
        IOobject::AUTO_WRITE
    ),
    mesh,
    dimensionedScalar(dimViscosity, Zero)
);
```

Temperature “T” field is created

Diffusivity as a field created

solver file ➔ diffusionFoam/diffusionFoam.C

```
#include "fvCFD.H"
#include "fvOptions.H"
#include "simpleControl.H"

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

int main(int argc, char *argv[])
{
    argList::addNote
    (
        "Laplace equation solver for a scalar quantity."
    );

    #include "postProcess.H"

    #include "addCheckCaseOptions.H"
    #include "setRootCaseLists.H"
    #include "createTime.H"
    #include "createMesh.H"

    simpleControl simple(mesh);

    #include "createFields.H"

    // * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

    Info<< "\nCalculating temperature distribution\n" << endl;

    while (simple.loop())
    {
        Info<< "Time = " << runTime.timeName() << nl << endl;

        while (simple.correctNonOrthogonal())
        {
            fvScalarMatrix TEqn
            (
                fvm::ddt(T) - fvm::laplacian(DT, T)
             ==
                fvOptions(T)
            );
```

Necessary “header” files

Matrix is created to solve “FVM”

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

```
        while (simple.correctNonOrthogonal())
        {
            fvScalarMatrix TEqn
            (
                fvm::ddt(T) - fvm::laplacian(DT, T)
             ==
                fvOptions(T)
            );

            fvOptions.constrain(TEqn);
            TEqn.solve();
            fvOptions.correct(T);
        }

        //#include "write.H"
        runTime.write();

        runTime.printExecutionTime(Info);
    }

    Info<< "End\n" << endl;

    return 0;
}
```

Compile diffusionFoam “solver”

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

```
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$ ls
CASE1_diffusionFOAM  diffusionFoam
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$ cd diffusionFoam/
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/diffusionFoam$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/diffusionFoam$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/diffusionFoam$ wmake
Making dependencies: diffusionFoam.C
g++ -std=c++11 -m64 -pthread -DOPENFOAM=2306 -DWM_DP -DWM_LABEL_SIZE=32 -Wall -Wextra -Wold-style-cast -Wnon-virtual-dtor -Wno-unused-parameter -Wno-invalid-offsetof -Wno-attributes -Wno-unknown-pragmas -
O3  -DNoRepository -ftemplate-depth-100  -I/home/openfoam/OpenFOAM/openfoam/src/finiteVolume/lnInclude -I/home/openfoam/OpenFOAM/openfoam/src/meshTools/lnInclude -iquote. -IlnInclude -I/home/openfoam/Open
FOAM/openfoam/src/OpenFOAM/lnInclude -I/home/openfoam/OpenFOAM/openfoam/src/OSspecific/POSIX/lnInclude   -fPIC -c diffusionFoam.C -o Make/linux64GccDPInt32Opt/diffusionFoam.o
g++ -std=c++11 -m64 -pthread -DOPENFOAM=2306 -DWM_DP -DWM_LABEL_SIZE=32 -Wall -Wextra -Wold-style-cast -Wnon-virtual-dtor -Wno-unused-parameter -Wno-invalid-offsetof -Wno-attributes -Wno-unknown-pragmas -
O3  -DNoRepository -ftemplate-depth-100  -I/home/openfoam/OpenFOAM/openfoam/src/finiteVolume/lnInclude -I/home/openfoam/OpenFOAM/openfoam/src/meshTools/lnInclude -iquote. -IlnInclude -I/home/openfoam/Open
FOAM/openfoam/src/OpenFOAM/lnInclude -I/home/openfoam/OpenFOAM/openfoam/src/OSspecific/POSIX/lnInclude   -fPIC -Xlinker --add-needed -Xlinker --no-as-needed  Make/linux64GccDPInt32Opt/diffusionFoam.o -L/h
ome/openfoam/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/lib \
    -lfiniteVolume -lfvOptions -lmeshTools -lOpenFOAM -ldl  \
    -lm -o /home/openfoam/OpenFOAM/openfoam-v2306/platforms/linux64GccDPInt32Opt/bin/diffusionFoam
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/diffusionFoam$
```

Compiled “diffusionFoam“ solver

case file

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

ComputationalThermalEngineering / DAY7-OpenFOAM_diffusion_equation / **CASE1_diffusionFOAM** /

Kumaresh0402 Delete DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM/system/as

Name		Last commit message
..		
0	Initial conditions	Delete DAY7-OpenFOAM_diffu
constant	Properties	Delete DAY7-OpenFOAM_diffu
system	Mesh, schemes, solvers, algorithm controls and tolerances for the implicit solvers.	Delete DAY7-OpenFOAM_diffu

Compile the “case” file named – **CASE1_diffusionFOAM**

```
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/diffusionFoam$ cd ..
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$ ls
CASE1_diffusionFOAM  diffusionFoam
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation$ cd CASE1_diffusionFOAM/
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$ blockMesh
/*---------------------------------------------------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  2306                                  |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
Build  : a6e826bd55-20230630 OPENFOAM=2306 version=v2306
Arch   : "LSB;label=32;scalar=64"
Exec   : blockMesh
Date   : Oct 23 2023
Time   : 16:04:30
Host   : openfoam
PID    : 80174
I/O    : uncollated
Case   : /home/openfoam/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM
nProcs : 1
trapFpe: Floating point exception trapping enabled (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 5, maxFileModificationPolls 20)
allowSystemOperations : Allowing user-supplied system call operations

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
Create time

Creating block mesh from "system/blockMeshDict"
Creating block edges
No non-planar block faces defined
Creating topology blocks

Creating topology patches - from boundary section
```

1

```
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$ diffusionFoam
/*---------------------------------------------------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  2306                                  |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
Build  : a6e826bd55-20230630 OPENFOAM=2306 version=v2306
Arch   : "LSB;label=32;scalar=64"
Exec   : diffusionFoam
Date   : Oct 23 2023
Time   : 16:06:31
Host   : openfoam
PID    : 80220
I/O    : uncollated
Case   : /home/openfoam/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM
nProcs : 1
trapFpe: Floating point exception trapping enabled (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 5, maxFileModificationPolls 20)
allowSystemOperations : Allowing user-supplied system call operations

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
Create time

Create mesh for time = 0


SIMPLE: no convergence criteria found. Calculations will run for 0.1 steps.

Reading field T

Reading diffusivity DT

No finite volume options present

Calculating temperature distribution

Time = 0.001

DICPCG:  Solving for T, Initial residual = 1, Final residual = 1.90148e-16, No Iterations 1
DICPCG:  Solving for T, Initial residual = 3.21798e-15, Final residual = 3.21798e-15, No Iterations 0
```

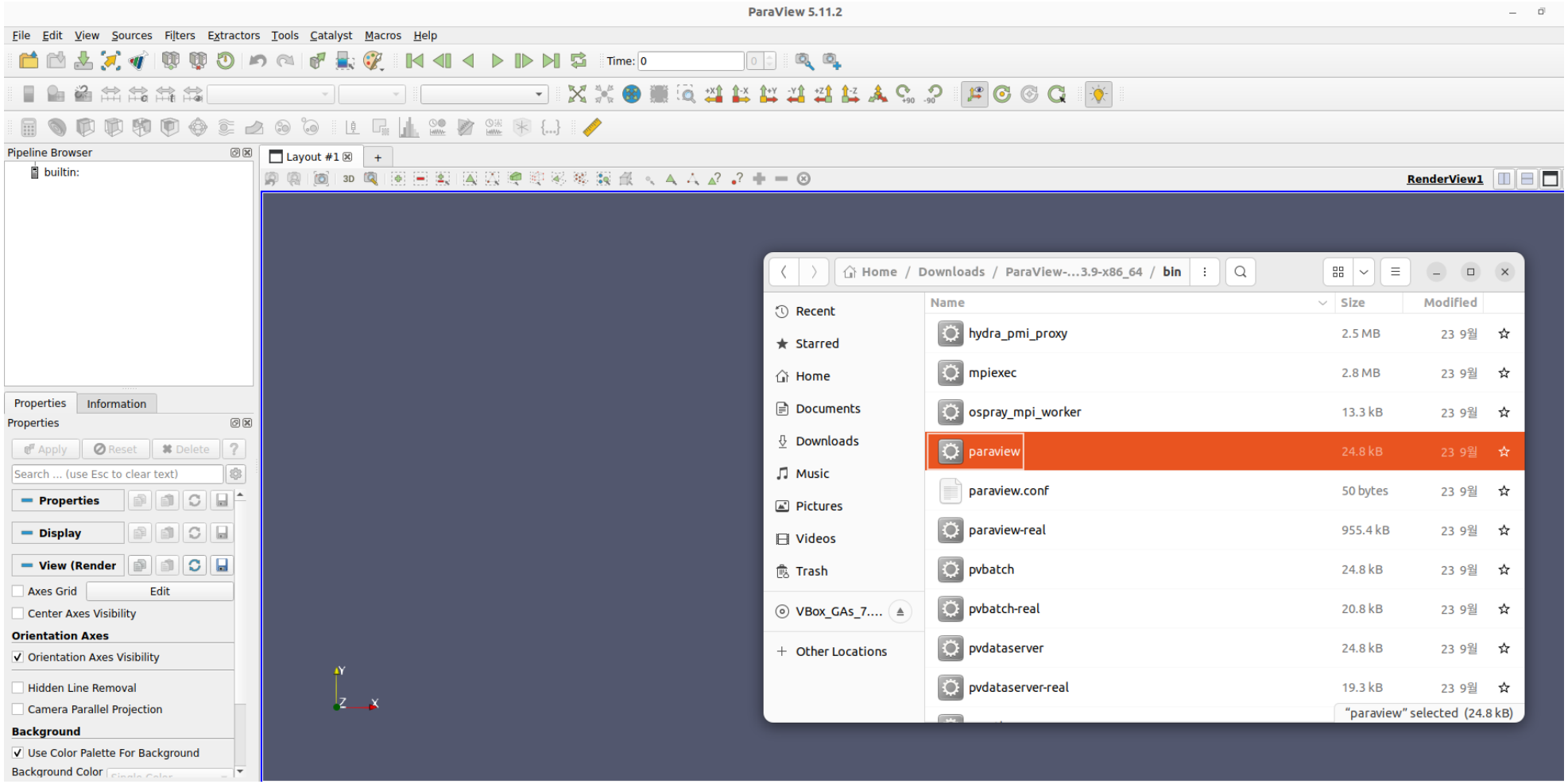
2

Compile the “case” file named – **CASE1_diffusionFOAM**

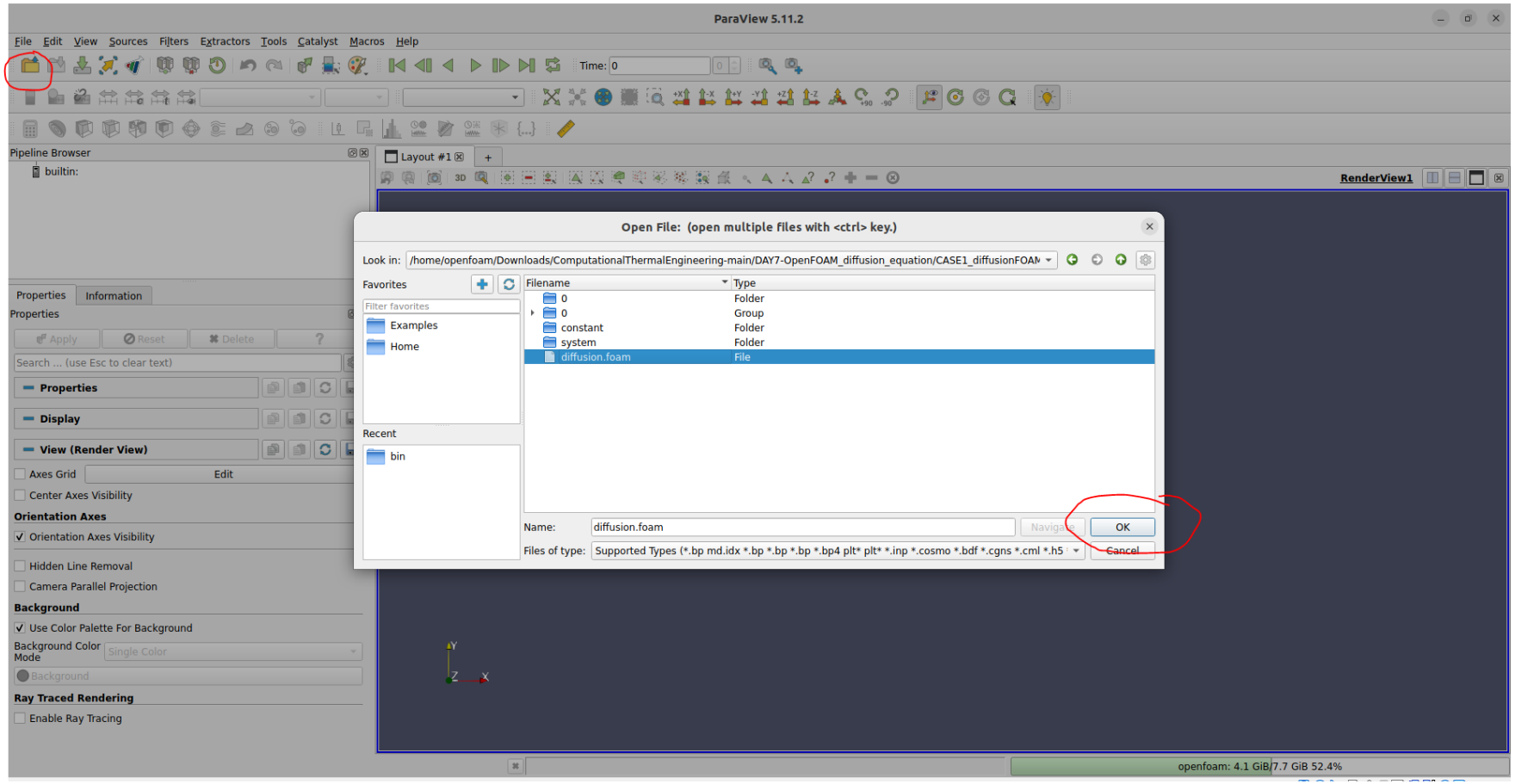
```
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$ ls
0      0.004  0.008  0.012  0.016  0.02   0.024  0.028  0.032  0.036  0.04   0.044  0.048  0.052  0.056  0.06   0.064  0.068  0.072  0.076  0.08   0.084  0.088  0.092  0.096  0.1
0.001  0.005  0.009  0.013  0.017  0.021  0.025  0.029  0.033  0.037  0.041  0.045  0.049  0.053  0.057  0.061  0.065  0.069  0.073  0.077  0.081  0.085  0.089  0.093  0.097  constant
0.002  0.006  0.01   0.014  0.018  0.022  0.026  0.03   0.034  0.038  0.042  0.046  0.05   0.054  0.058  0.062  0.066  0.07   0.074  0.078  0.082  0.086  0.09   0.094  0.098  system
0.003  0.007  0.011  0.015  0.019  0.023  0.027  0.031  0.035  0.039  0.043  0.047  0.051  0.055  0.059  0.063  0.067  0.071  0.075  0.079  0.083  0.087  0.091  0.095  0.099
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$ touch diffusion.foam
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$ ls
0      0.004  0.008  0.012  0.016  0.02   0.024  0.028  0.032  0.036  0.04   0.044  0.048  0.052  0.056  0.06   0.064  0.068  0.072  0.076  0.08   0.084  0.088  0.092  0.096  0.1
0.001  0.005  0.009  0.013  0.017  0.021  0.025  0.029  0.033  0.037  0.041  0.045  0.049  0.053  0.057  0.061  0.065  0.069  0.073  0.077  0.081  0.085  0.089  0.093  0.097  constant
0.002  0.006  0.01   0.014  0.018  0.022  0.026  0.03   0.034  0.038  0.042  0.046  0.05   0.054  0.058  0.062  0.066  0.07   0.074  0.078  0.082  0.086  0.09   0.094  0.098  diffusion.foam
0.003  0.007  0.011  0.015  0.019  0.023  0.027  0.031  0.035  0.039  0.043  0.047  0.051  0.055  0.059  0.063  0.067  0.071  0.075  0.079  0.083  0.087  0.091  0.095  0.099  system
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
openfoam@openfoam:~/Downloads/ComputationalThermalEngineering-main/DAY7-OpenFOAM_diffusion_equation/CASE1_diffusionFOAM$
```

3

Open “Paraview” ➡ Used for post-processing your results



Open the “case” in paraview



4

Project 1 - Solving diffusion equation in OpenFOAM #7

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Kumaresh1176 started this conversation in **General**

Kumaresh1176 last week Maintainer

Based on DAY 6 presentation, repeat all steps we discussed during the session

> Make sure OpenFOAM is installed on your systems.
> Install ParaView.
> Copy solver and test case to the working directory.
> Build/compile the solver.

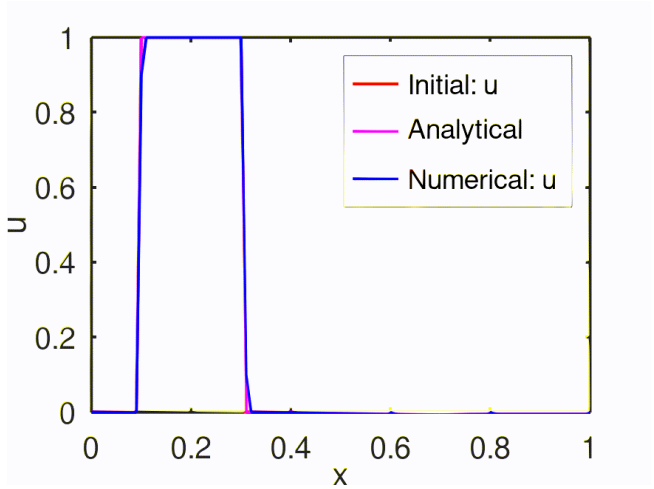
1. wclean
2. wmake

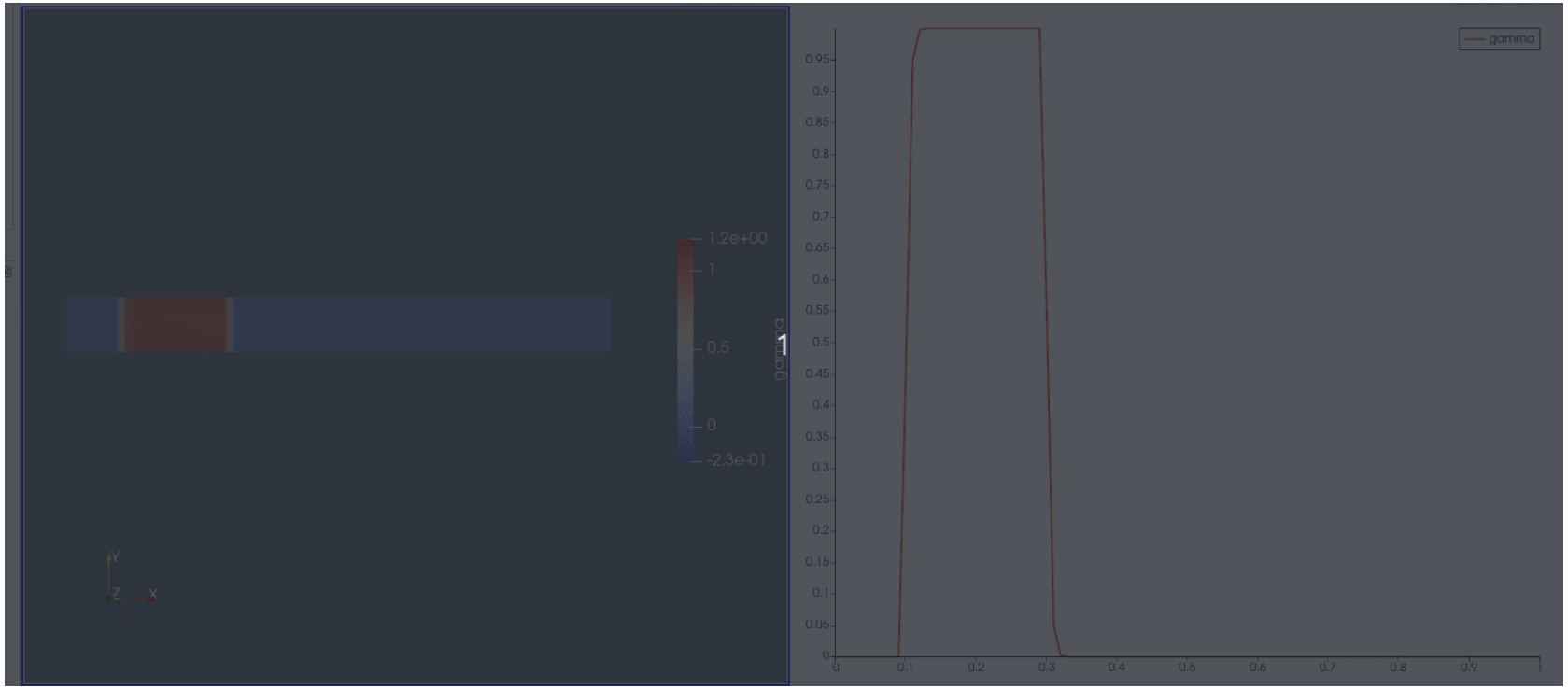
> Run the test case by using following commands:

1. blockMesh
2. diffusionFoam
3. touch a.foam

> Visualize the results.
> Share screenshots of results here with clear description

OpenFOAM: Numerical Solution to Convection Equation



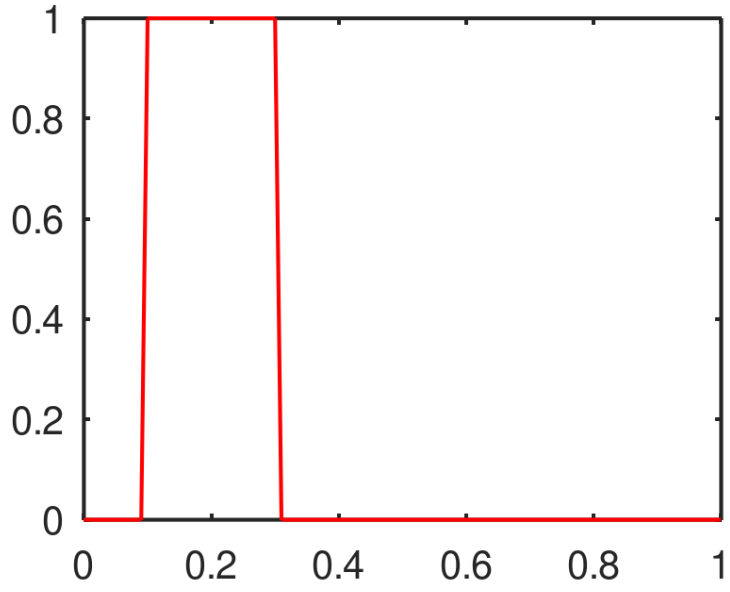


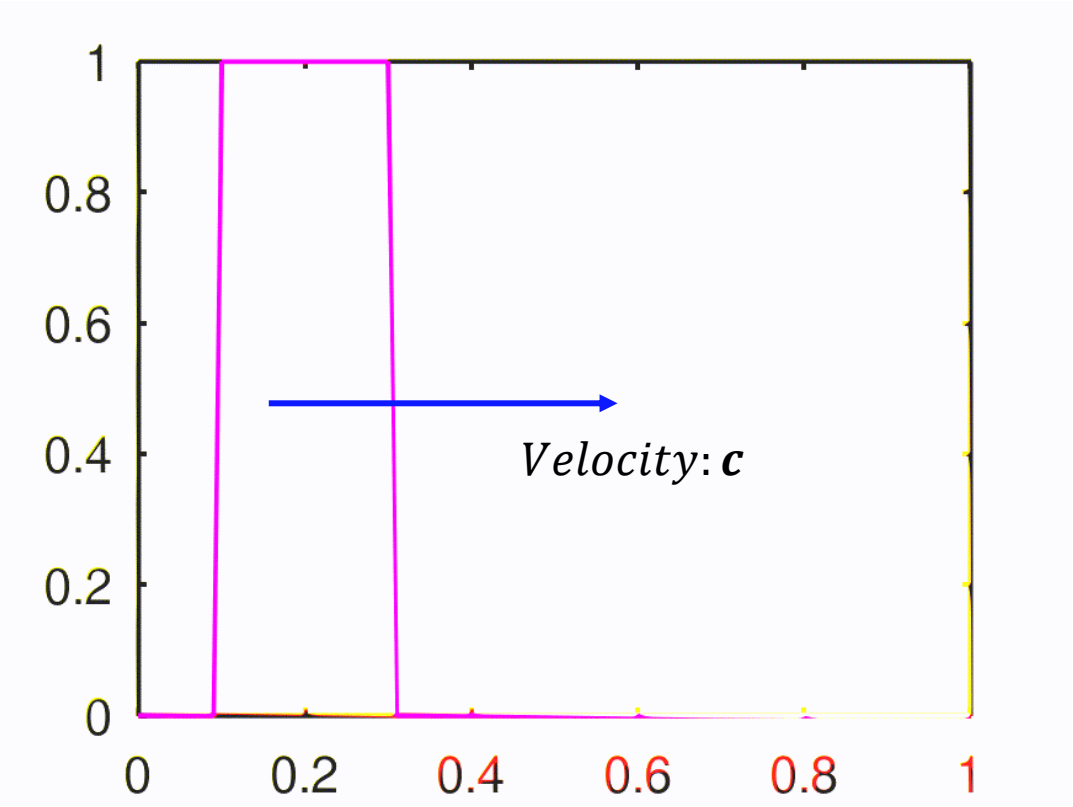
$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

Convection Equation

Setting initial field

```
for i = 1 : length(x)
  if (x(i, 1) >= 0.1) && (x(i, 1) <= 0.3)
    u(i, 1) = 1;
  endif
end
```

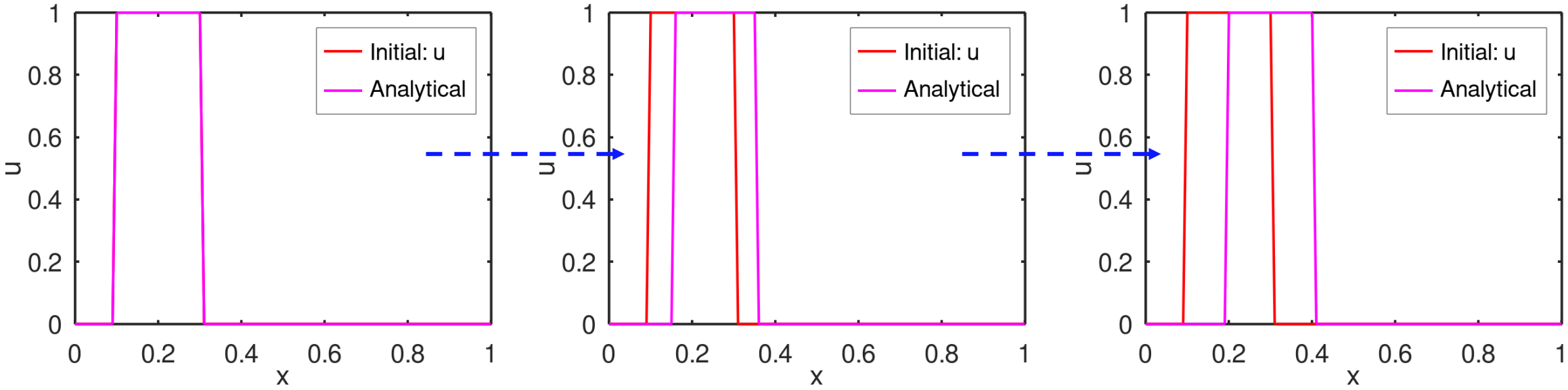




Convection Equation

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$ ← Advection equation

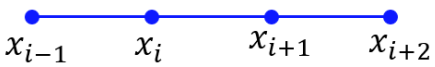
```
for i = 1 : length(x)
  if (x(i, 1) >= 0.1+c*t) && (x(i, 1) <= 0.3+c*t)
    u_analytical(i, 1) = 1;
  endif
end
```



Convection Equation

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + c\left(\frac{\partial u}{\partial x}\right)_i^n = 0$$



$$\left(\frac{d\rho}{dx}\right)_i \approx \frac{\rho(x_{i+1}) - \rho(x_i)}{\Delta x_i}$$

$$\left(\frac{d\rho}{dx}\right)_i \approx \frac{\rho(x_{i+1}) - \rho(x_{i-1})}{2\Delta x_i}$$

$$u_i^{n+1} = u_i^n - c\Delta t\left(\frac{\partial u}{\partial x}\right)_i^n$$

$$\left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_{i+1}^n - u_i^n}{\Delta x_i}$$

Simple forward difference scheme

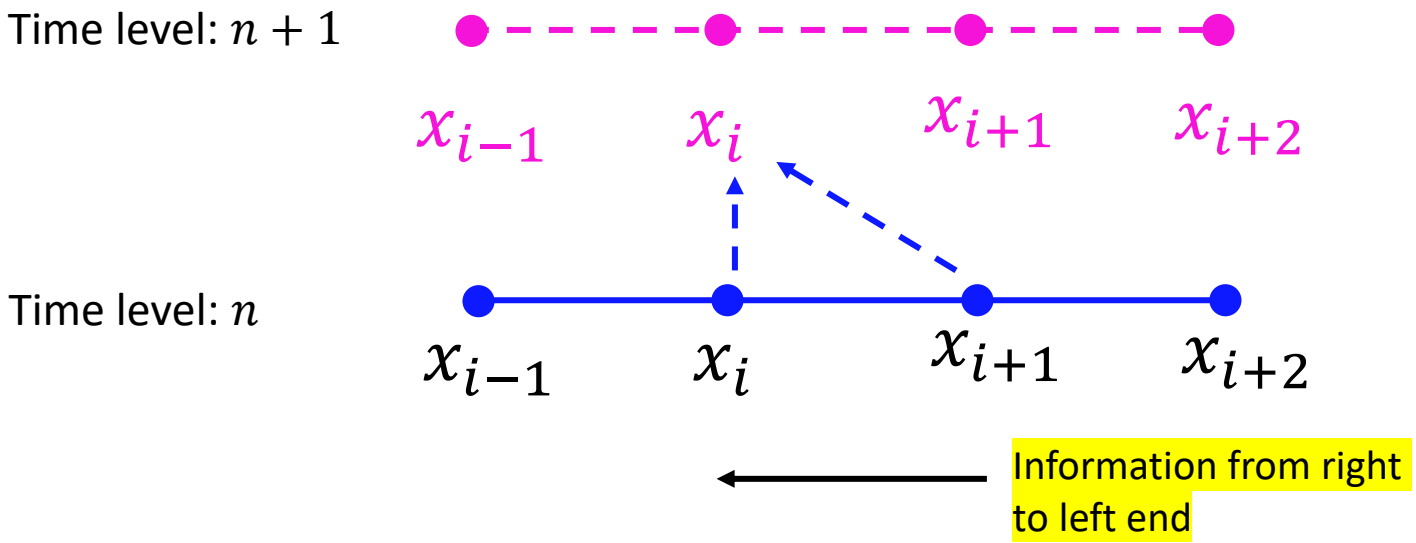
$$\left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x_i}$$

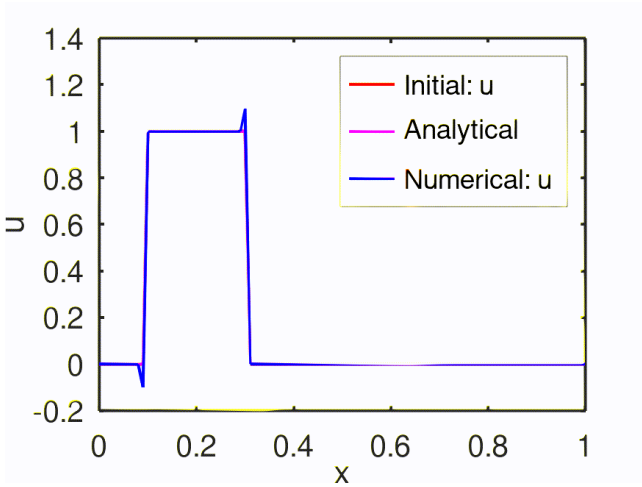
Central difference

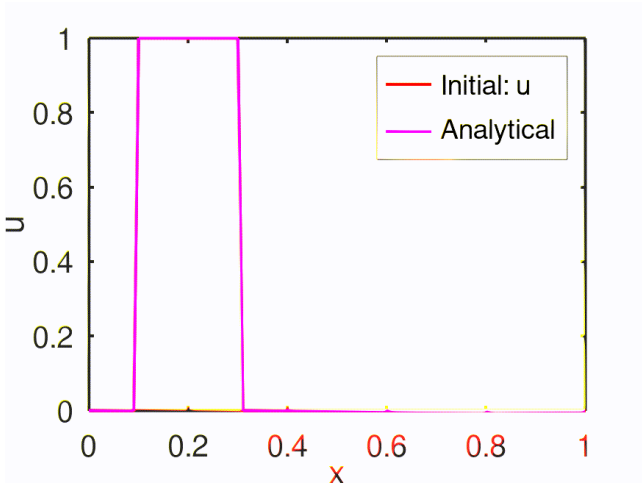
(Explicit) First order - Forward Euler

(only one unknown (n+1) with other knowns at n^{th} node)
→ conditionally stable

Convection Equation







$$u_i^{n+1} = u_i^n - c\Delta t \left(\frac{\partial u}{\partial x}\right)_i^n$$

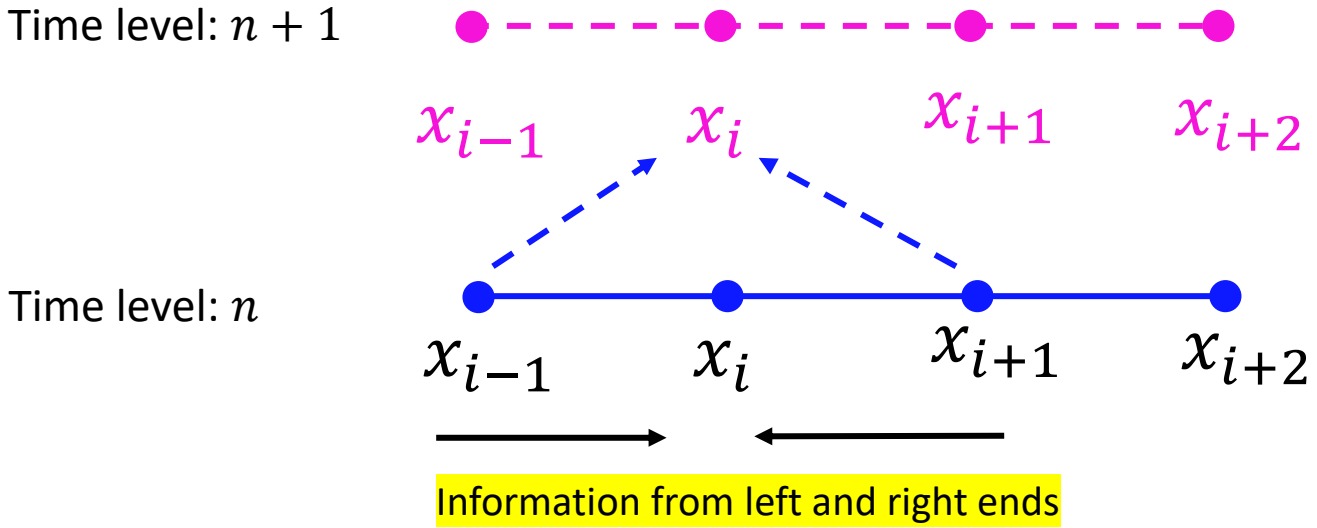
$$\left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_{i+1}^n - u_i^n}{\Delta x_i}$$

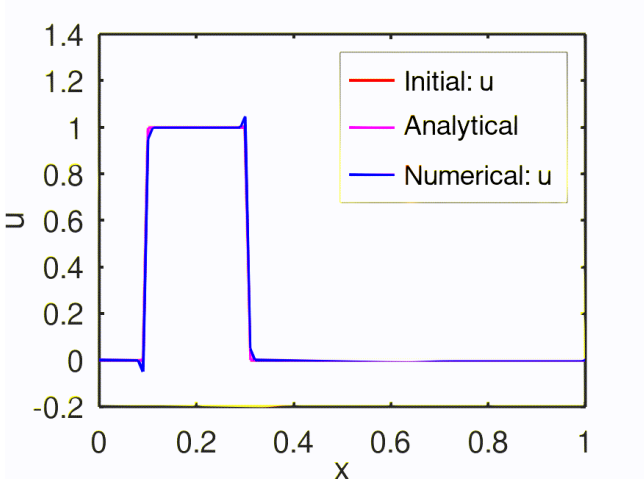
Simple forward difference scheme

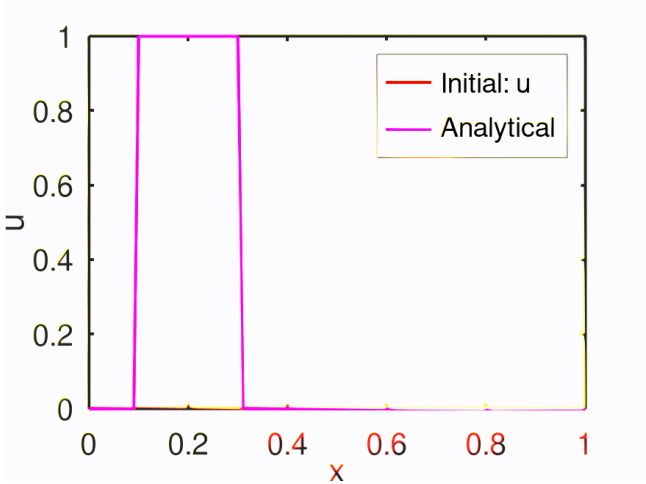
$$\left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x_i}$$

Central difference

Convection Equation







$$u_i^{n+1} = u_i^n - c\Delta t \left(\frac{\partial u}{\partial x}\right)_i^n$$

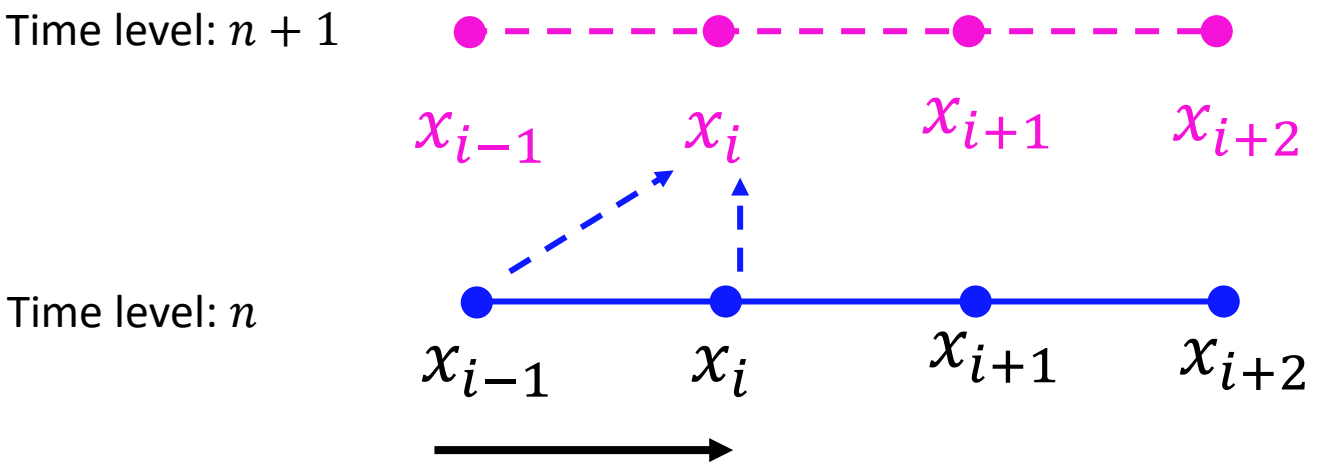
$$\left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_{i+1}^n - u_i^n}{\Delta x_i}$$

Simple forward difference scheme

$$\left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x_i}$$

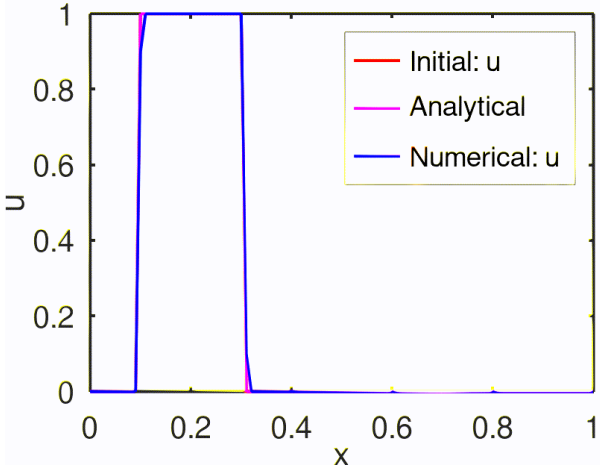
Central difference

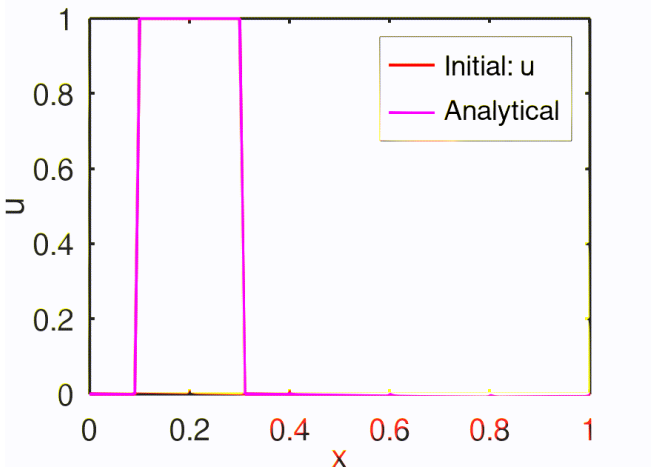
Convection Equation



Information from left to right end

Wind is flowing from left end (bird moves from left to right)





$$u_i^{n+1} = u_i^n - c\Delta t \left(\frac{\partial u}{\partial x}\right)_i^n \longrightarrow \left(\frac{\partial u}{\partial x}\right)_i^n \approx \frac{u_i^n - u_{i-1}^n}{\Delta x_i}$$

Simple backward difference scheme

CFL = 0.1

$$CFL: \frac{c\Delta t}{\Delta x}$$

CFL < 1 → Numerically stable (conditionally stable based on the condition imposed by CFL) – EXPLICIT method

CFL > = 1 → Numerically unstable

Project 2 - Solving convection equation in OpenFOAM #8

Kumaresh1176 started this conversation in **General**

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

Kumaresh1176 15 minutes ago Maintainer

Based on DAY 7 presentation, repeat all steps we discussed during the session

> Make sure OpenFOAM is installed on your systems.
> Install ParaView.
> Copy solver and test case to the working directory.
> Build/compile the solver.

1. wclean
2. wmake

> Run the test case by using following commands:
>
> > Command to clean: foamCleanTutorials

1. blockMesh
2. setFields
3. simpleConvection
4. touch a.foam

> Visualize the results.
> Share screenshots of results here with clear description